

# Laboratório 7

ra: 175886

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1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(DBI)
```

Warning: package 'DBI' was built under R version 4.5.3

```
library(RSQLite)
```

Warning: package 'RSQLite' was built under R version 4.5.3

```
conn <- dbConnect(RSQLite::SQLite(), "C:\\Users\\jvsca\\Desktop\\Unicamp\\Banco de Dados\\
```

```
dbListTables(conn)
```

```
[1] "course_offerings"  "courses"           "grade_distributions"
[4] "instructors"       "rooms"              "schedules"
[7] "sections"          "subject_memberships" "subjects"
[10] "teachings"
```

```
apply(dbListTables(conn), function(tb) dbListFields(conn, tb))
```

```
$course_offerings
```

```
[1] "uuid"           "course_uuid" "term_code"    "name"
```

```

$courses
[1] "uuid"    "name"    "number"

$grade_distributions
[1] "course_offering_uuid" "section_number" "n_count"
[4] "c_count"             "f_count"         "cr_count"
[7] "p_count"             "nw_count"        "other_count"
[10] "bc_count"            "u_count"         "d_count"
[13] "ab_count"            "a_count"         "s_count"
[16] "i_count"             "b_count"         "nr_count"

$instructors
[1] "id"    "name"

$rooms
[1] "uuid"          "facility_code" "room_code"

$schedules
[1] "uuid"          "start_time" "end_time" "mon"      "tues"
[6] "wed"           "thurs"      "fri"      "sat"      "sun"

$sections
[1] "uuid"          "course_offering_uuid" "section_type"
[4] "number"        "room_uuid"           "schedule_uuid"

$subject_memberships
[1] "subject_code"    "course_offering_uuid"

$subjects
[1] "code"          "name"          "abbreviation"

$teachings
[1] "instructor_id" "section_uuid"

```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT')

```

SELECT DISTINCT i.id, i.name
FROM instructors AS i
INNER JOIN teachings AS t ON t.instructor_id = i.id
INNER JOIN sections AS se ON se.uuid = t.section_uuid
INNER JOIN course_offerings AS co ON co.uuid = se.course_offering_uuid

```

```
INNER JOIN subject_memberships AS sm ON sm.course_offering_uuid = co.uuid
INNER JOIN subjects AS s ON s.code = sm.subject_code
WHERE s.abbreviation = 'STAT'
```

Table 1: Displaying records 1 - 10

| id      | name            |
|---------|-----------------|
| 4146030 | SHENG WANG      |
| 393380  | DONALD PORTER   |
| 4392906 | MINJING TAO     |
| 5697552 | KUNLING HUANG   |
| 4636315 | JARED BROWN     |
| 4898014 | DONG XIA        |
| 4613728 | JENNIFER NGUYEN |
| 5700058 | YUCHEN ZHOU     |
| 5520811 | SEULKEE YUN     |
| 5696436 | YONGFENG WU     |

Número exato de professores

```
SELECT COUNT(DISTINCT i.id) AS total_professores
FROM instructors AS i
INNER JOIN teachings AS t ON t.instructor_id = i.id
INNER JOIN sections AS se ON se.uuid = t.section_uuid
INNER JOIN course_offerings AS co ON co.uuid = se.course_offering_uuid
INNER JOIN subject_memberships AS sm ON sm.course_offering_uuid = co.uuid
INNER JOIN subjects AS s ON s.code = sm.subject_code
WHERE s.abbreviation = 'STAT'
```

Table 2: 1 records

| total_professores |
|-------------------|
| 379               |

3. O GPA americano é definido numa escala de 0 a 4, em que  $A = 4$ ,  $AB = 3.5$ ,  $B = 3$ ,  $BC = 2.5$ ,  $C = 2$ ,  $D = 1$  e  $F = 0$ . Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

a) Quem é o professor mais difícil?

- b) Quem é o professor mais fácil?
- c) Qual é a disciplina mais difícil?
- d) Qual é a disciplina mais fácil?

Quem é o professor mais difícil?

```
SELECT
    i.name AS professor_mais_dificil,
    SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.5) AS avg_gpa,
    CAST(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count) AS DECIMAL(10,4)) AS total_count
FROM grade_distributions gd
JOIN sections sec ON gd.course_offering_uuid = sec.course_offering_uuid AND gd.section_number = sec.section_number
JOIN teachings t ON sec.uuid = t.section_uuid
JOIN instructors i ON t.instructor_id = i.id
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
    AND (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.e_count) > 0
GROUP BY i.name
ORDER BY avg_gpa ASC
LIMIT 1
```

Table 3: 1 records

| professor_mais_dificil | avg_gpa |
|------------------------|---------|
| JAMES D KUELBS         | 2.6375  |

- b) Quem é o professor mais fácil?

```
SELECT
    i.name AS professor_mais_facil,
    SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.5) AS avg_gpa,
    CAST(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count) AS DECIMAL(10,4)) AS total_count
FROM grade_distributions gd
JOIN sections sec ON gd.course_offering_uuid = sec.course_offering_uuid AND gd.section_number = sec.section_number
JOIN teachings t ON sec.uuid = t.section_uuid
JOIN instructors i ON t.instructor_id = i.id
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
    AND (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.e_count) > 0
GROUP BY i.name
ORDER BY avg_gpa DESC
LIMIT 1
```

```
ORDER BY avg_gpa DESC
LIMIT 1
```

Table 4: 1 records

| professor_mais_facil | avg_gpa |
|----------------------|---------|
| YAJUAN SI            | 4       |

c) Qual é a disciplina mais difícil?

```
SELECT
  c.name AS disciplina_mais_dificil,
  SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) AS avg_gpa,
  CAST(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count) AS DECIMAL(10,4)) AS total_count
FROM grade_distributions gd
JOIN course_offerings co ON gd.course_offering_uuid = co.uuid
JOIN courses c ON co.course_uuid = c.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
  AND (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.e_count + gd.f_count + gd.g_count + gd.h_count + gd.i_count + gd.j_count + gd.k_count + gd.l_count + gd.m_count + gd.n_count + gd.o_count + gd.p_count + gd.q_count + gd.r_count + gd.s_count + gd.t_count + gd.u_count + gd.v_count + gd.w_count + gd.x_count + gd.y_count + gd.z_count) > 0
GROUP BY c.name
ORDER BY avg_gpa ASC
LIMIT 1;
```

Table 5: 1 records

| disciplina_mais_dificil                   | avg_gpa  |
|-------------------------------------------|----------|
| Introduction to the Theory of Probability | 2.886374 |

d) Qual é a disciplina mais fácil?

```
SELECT
  c.name AS disciplina_mais_facil,
  SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) AS avg_gpa,
  CAST(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count) AS DECIMAL(10,4)) AS total_count
FROM grade_distributions gd
JOIN course_offerings co ON gd.course_offering_uuid = co.uuid
JOIN courses c ON co.course_uuid = c.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
```

```

JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
      AND (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.
GROUP BY c.name
ORDER BY avg_gpa DESC
LIMIT 1

```

Table 6: 1 records

| disciplina_mais_facil           | avg_gpa |
|---------------------------------|---------|
| Sample Survey Theory and Method | 4       |

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-23 20:48:33 -03"
```